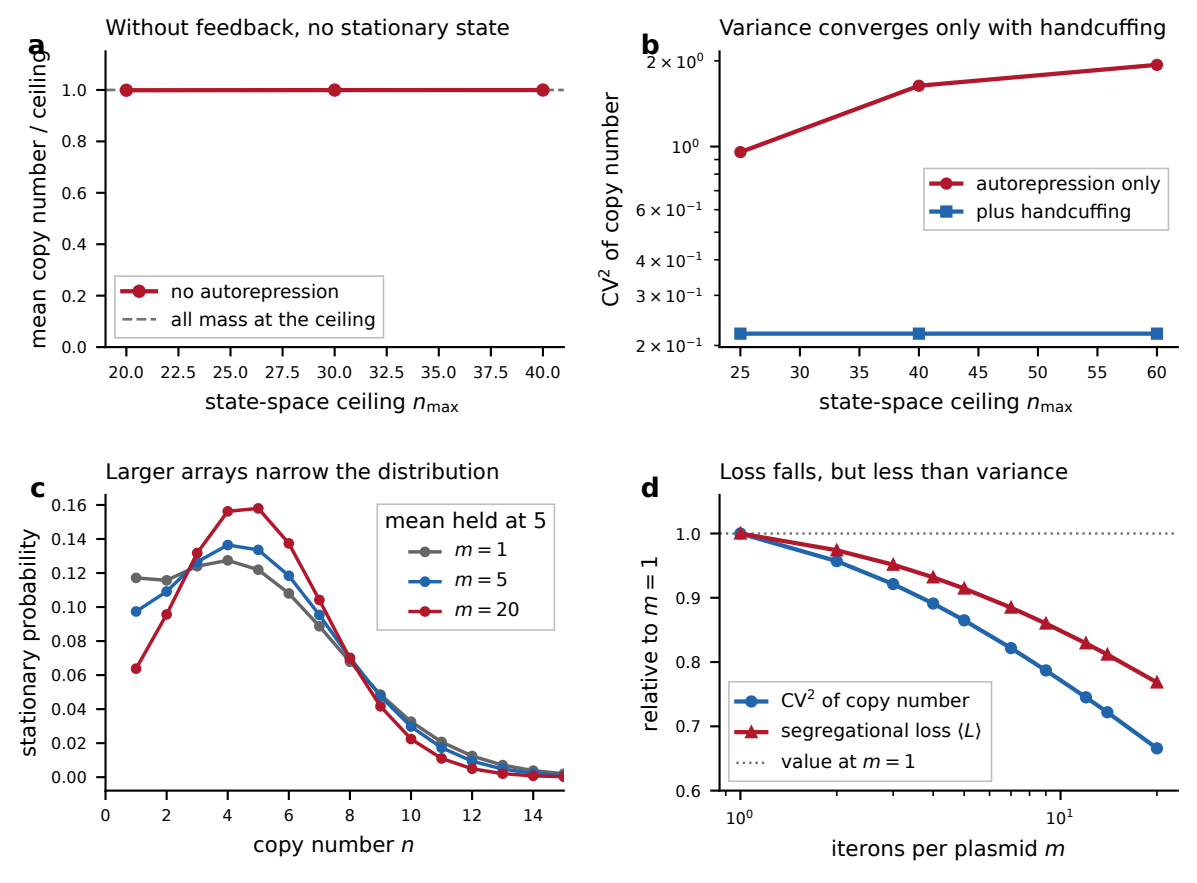
a
Without feedback, no stationary state
mean copy number / ceiling
1.0
0.8
0.6
0.4
0.2
0.0
no autorepression
all mass at the ceiling
20.0 22.5 25.0 27.5 30.0 32.5 35.0 37.5 40.0
state-space ceiling n_{max}
b
Variance converges only with handcuffing
CV^2 of copy number
2×10^0
10^0
6×10^{-1}
4×10^{-1}
3×10^{-1}
2×10^{-1}
autorepression only
plus handcuffing
25 30 35 40 45 50 55 60
state-space ceiling n_{max}
c
Larger arrays narrow the distribution
stationary probability
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
mean held at 5
$m = 1$
$m = 5$
$m = 20$
0 2 4 6 8 10 12 14
copy number n
d
Loss falls, but less than variance
relative to $m = 1$
1.0
0.9
0.8
0.7
0.6
CV^2 of copy number
segregational loss $\langle L \rangle$
value at $m = 1$
10^0
10^1
iterons per plasmid m